

MICRO-DEGREE

Künstliche Intelligenz und Gesellschaft

***Artificial Intelligence
and Society***



Micro-Degree Artificial Intelligence and Society

– Technical Aspects of AI –

Lecture 4.6: The Recommender System of X

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Research group

Complex Social & Computational Systems (CS)²

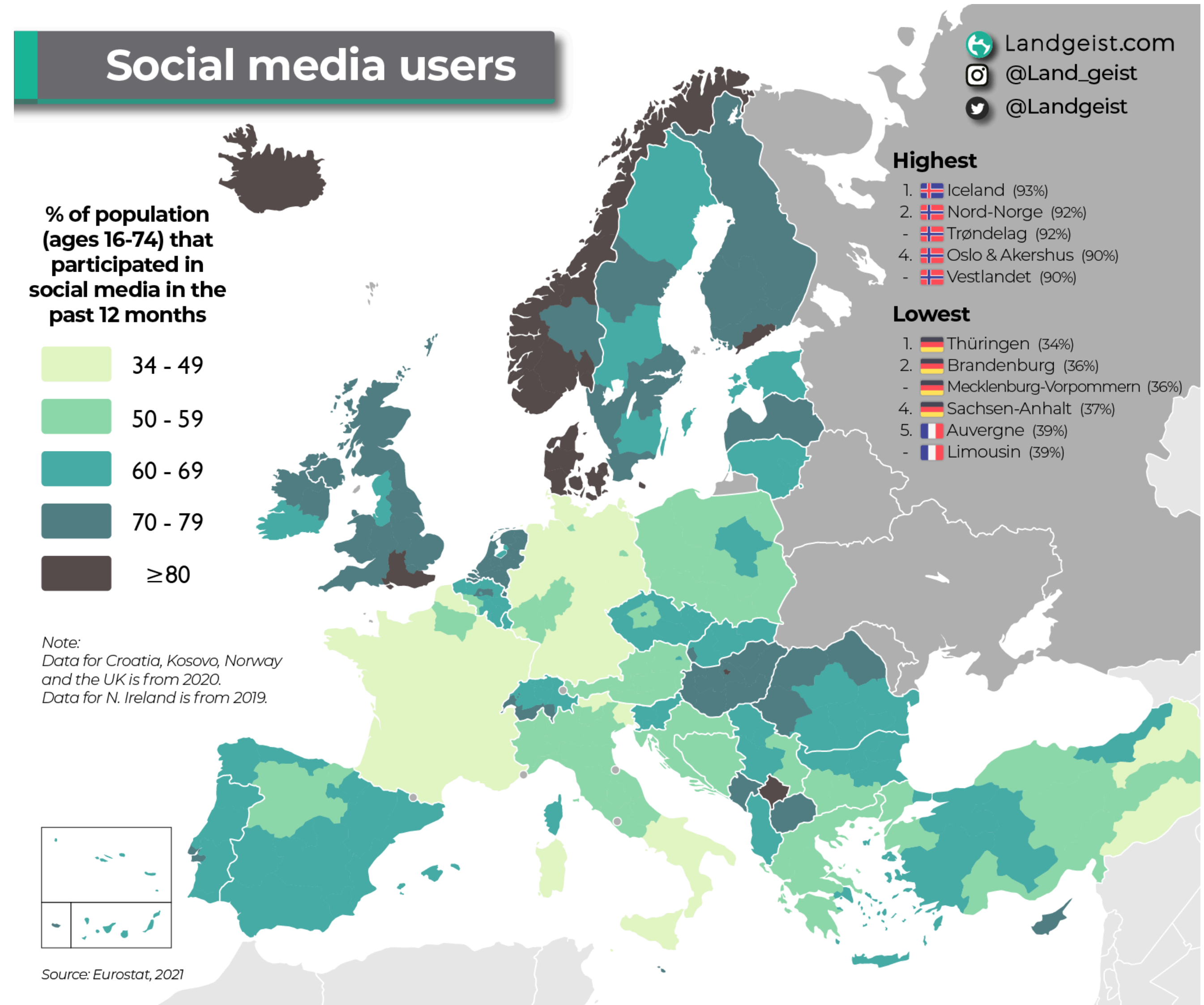
IDEa_Lab

Das Interdisziplinäre Digitale
Labor der Universität Graz



Social Media

- Large parts of the population in the EU use social media.
- Among 16-24 year olds, the proportion is already 74%.
- On a platform like X (formerly "Twitter"), about 500 million posts are created daily.
- However, a typical user only sees a few hundred of them.
- → All major social media platforms have complex recommender systems.



Source: [Landgeist.com](https://landgeist.com).

The Recommender System of X

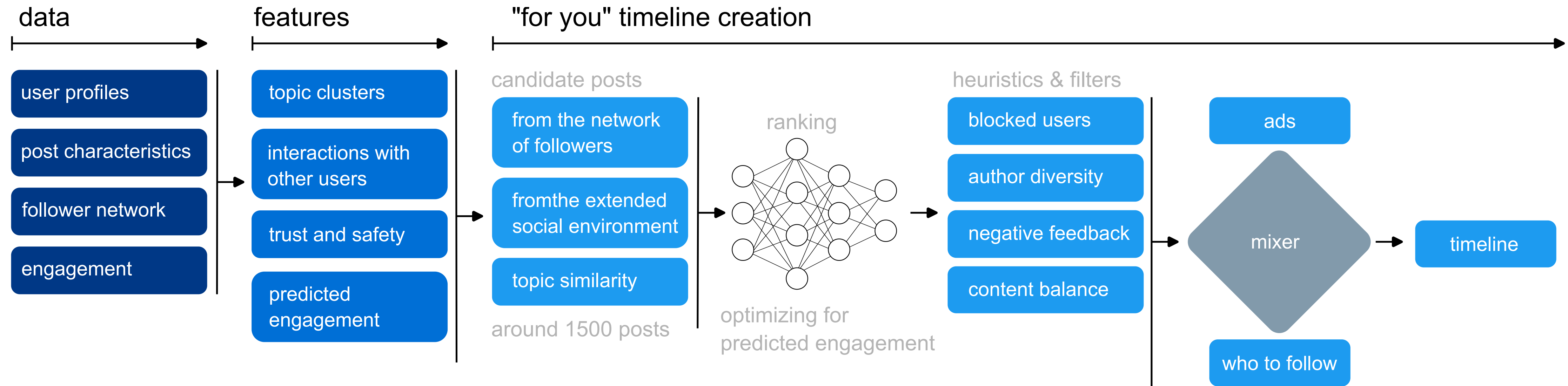
Tasks of the recommender system

1. To **select** posts that are relevant to the user.
2. To **arrange** posts in a timeline – the most relevant first.
3. To **detect illegal content** and keep it out of users' timelines.

Sources of information

1. **Information about users** such as their profile and stated interests.
2. **Information about posts** such as topic, multimedia content, and links.
3. **Information about the community** such as which other users a user follows, and which posts these users interacted with.
4. **Information about user behavior** such as which posts users liked, shared, or replied to in the past (engagement) → optimization goal.

From Data to Recommendations



- Text data is embedded to process it into features.
- Using the features, the cosine similarity to content & users the user had interactions with is calculated.
- For the 1500 most similar or "most relevant" posts, a neural network predicts how likely the user is to engage with them.
- After applying filters, the ranked posts are mixed with ads and follow recommendations and delivered as a timeline.

Problems with Optimizing Engagement

Strongly emotionally charged or sensational content (clickbait) excites us and leads us to engage more with it. This leads to ...



- ... extreme views spreading faster and wider [1, 2].



- ... declining social trust and increasing polarization [3].



- ... political literacy decreasing [4].

[1] Lorenz-Spreen et al., [A Systematic Review of Worldwide Causal and Correlational Evidence on Digital Media and Democracy](#), *Nat Hum Behav* (2021).

[2] Müller & Schwarz, [Fanning the Flames of Hate](#), *J Eur Econ Assoc* (2023).

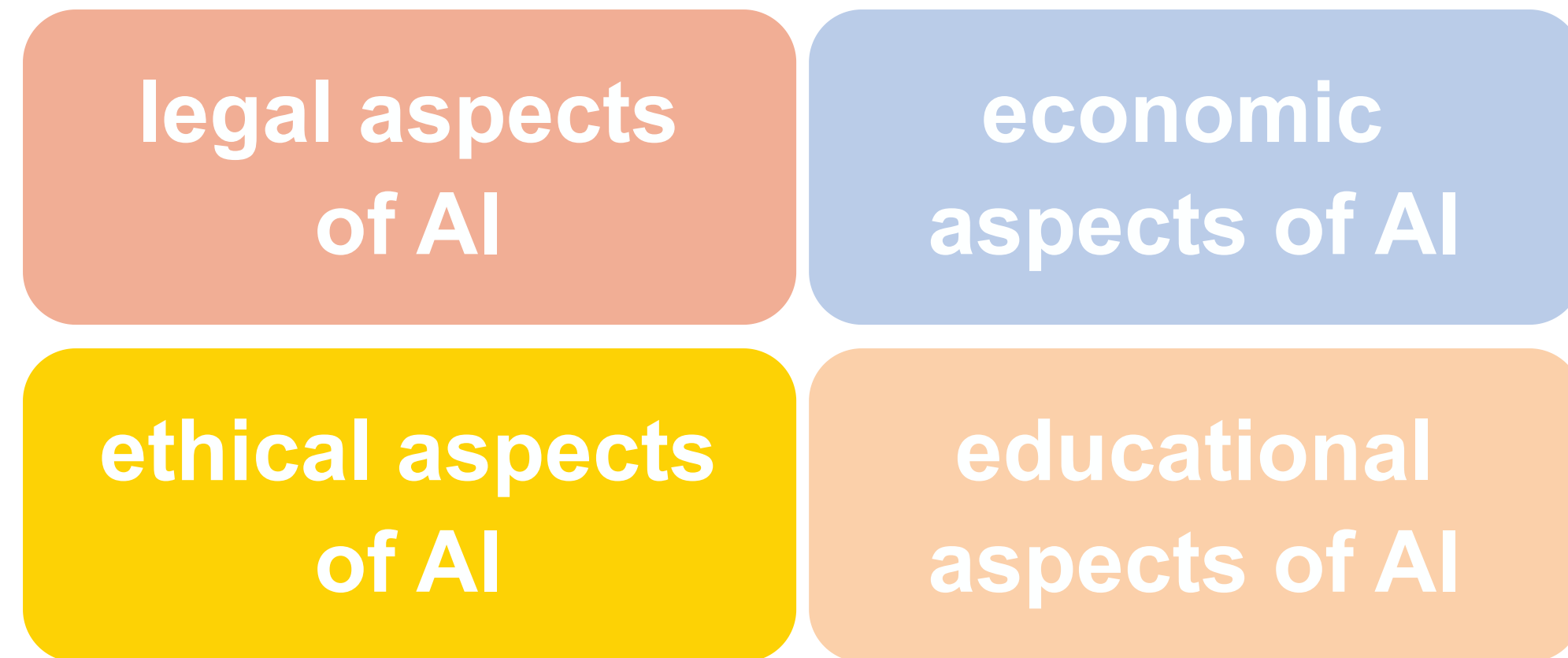
[3] Sabatini & Sarracino, [Online Social Networks and Trust](#), *Soc Indic Res* (2021).

[4] Lee, [Connecting Social Media Use with Gaps in Knowledge and Participation in a Protest Context](#), *Asian J Commun*, (2019).

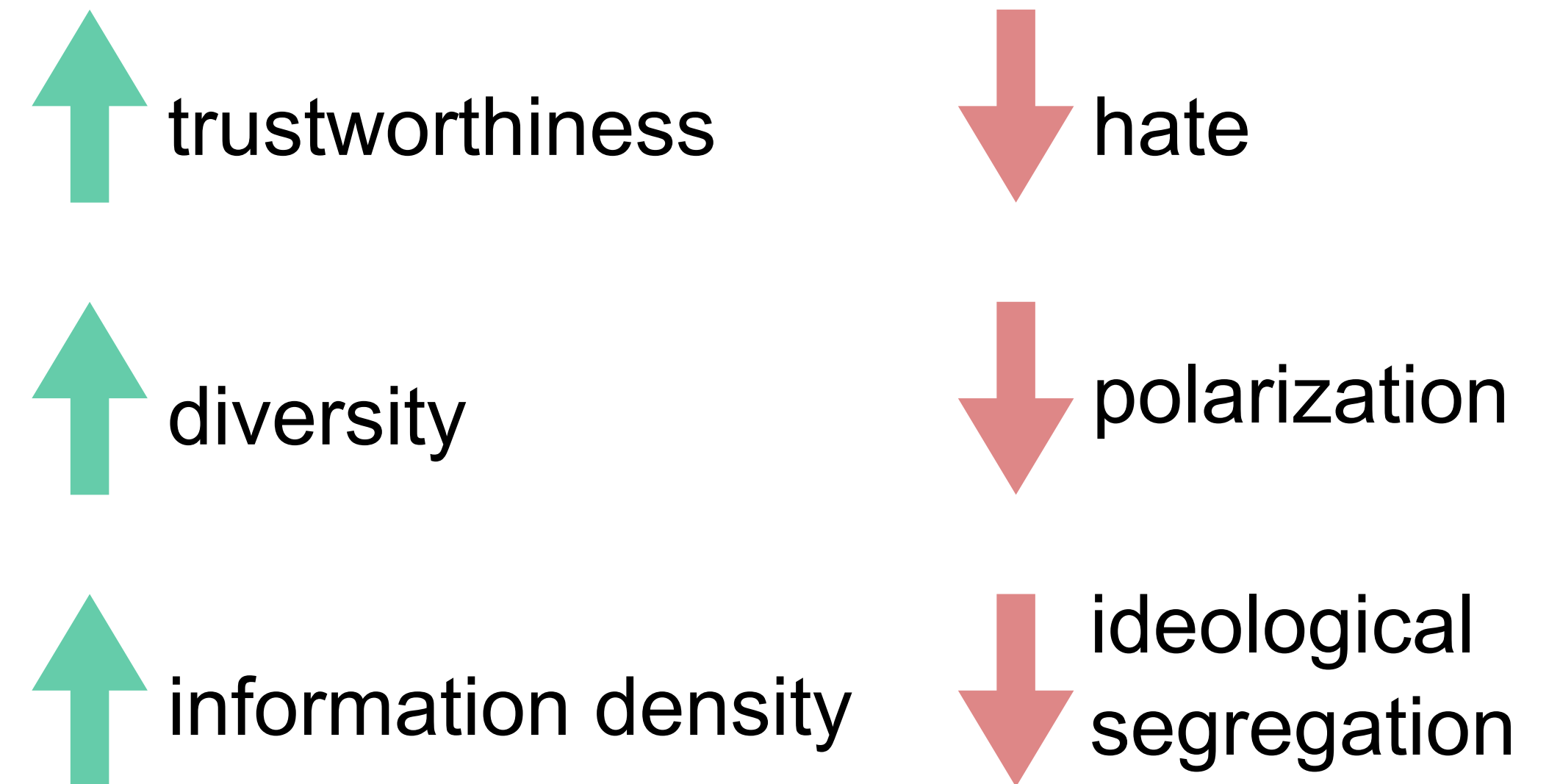
Reducing Societal Risks

Providers [...] shall diligently identify [...] any actual or foreseeable negative effects on **civic discourse** and **electoral processes**, and **public security** [...]

Digital Services Act, Article 34



Idea: New recommendation algorithms that not only optimize engagement but also pay attention to other characteristics.



Research project "Designing Social Media Recommendation Algorithms for Societal Good".

Summary

- Social media platforms like X (formerly Twitter) use complex recommender systems based on machine learning techniques to select content.
- These recommender systems optimize user engagement to maximize the platform's advertising revenue.
- Engagement optimization leads to societal problems which are addressed in laws like the Digital Services Act.

Technical Foundations of AI

- **Different Forms of Machine Learning**
 - Supervised Learning
 - Unsupervised Learning
 - Reinforcement Learning
- **Computational Thinking & Mathematical Foundations**
 - Deconstructing problems so they can be solved by algorithms
 - Measures for algorithm performance
 - Measures for the distance or similarity of data points
- **Data as the Basis of Artificial Intelligence**
 - Data types (numbers, categories, images, text)
 - Problems with data
 - Feature engineering and dimensionality reduction
- **Applications of Artificial Intelligence**
 - Classification, Regression, Cluster Analysis, dimensionality reduction.
 - Large Language Models (LLMs)
 - Recommender Systems

Where does "AI" work (not) well?

- **Huge Advances:** Optical character recognition, translation, image recognition, ...
→ **Perception**
- **Passable Advances:** Spam detection, content moderation, detection of financial crimes, ...
→ **Automation of decisions**
- **Very Questionable:** Prediction of criminal recidivism, prediction of terrorism, prediction of job performance ...
→ **Prediction of social characteristics**